02/01/2026 (FN)

Hall Ticket Number:

MR-22

Code: B325CS5

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Regular/Supplementary Examinations, December 2025/January 2026

Computer Organization and Architecture

(Common to CSE, CSB, CSM and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	What is the role of AR and DR registers?	1	1	1	1
1. b)	In which addressing mode, the actual operand is given in the instruction?	1	1	2	1
1. c)	What is control memory?	1	2	2	2
1. d)	Describe ASHL and SHRA operations.	1	2	2	2
1. e)	Convert the following numbers. i. $(9FD)_{16}=(_____)_8$ ii. $(234)_{10}=(_____)_{16}$	1	3	5	2
1. f)	Draw the hardware configuration for signed 2's complement addition and subtraction.	1	1	5	1
1. g)	What are different modes of data transfer?	1	1	3	1
1. h)	How EEPROM differs from PROM?	1	1	3	1
1. i)	List the characteristics of RISC architecture.	1	1	3	2
1. j)	Write the formula to calculate speedup of a pipelined processor over an equivalent non-pipeline processor.	1	1	4	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain the functionality of 4-bit combinational circuit shifter with a neat block diagram.	5	1	3	2
2. b)	What is an interrupt? Draw the flowchart of interrupt cycle?	5	1	2	2
	OR				
2. c)	Design a 4-bit Arthmetic circuit using full adders.	5	2	3	2
2. d)	Describe the following memory reference instructions with examples. i. AND ii. ADD iii. Load	5	2	2	2
3. a)	Explain microprogram sequencer for a control memory with a neat block diagram.	5	1	2	2
3. b)	Write the assembly language program to evaluate A= (B*C)/(D-E) using three address instruction format.	5	2	3	2
	OR				
3. c)	Explain the block diagram for micro programmed control unit.	5	1	2	2
3. d)	Explain the following addressing modes with examples i. Immediate mode ii. Direct addressing mode iii. Indexed addressing mode.	5	2	3	2
4. a)	Perform the following arithmetic operations. i. (+54)+(-21) ii. (-34)+(-54)	5	3	3	2
4. b)	Draw the flow chart for Booth multiplication algorithm and multiply 13 with -7 using Booth multiplication algorithm.	5	1	3	2

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	OR				
4. c)	Explain the flowchart of floating point addition/subtraction operation.	5	2	3	2
4. d)	Divide 13 with 7 using binary division.	5	2	3	2
5. a)	Compare Isolated I/O and Memory mapped I/O.	5	1	2	3
5. b)	Write short notes on i. ROM ii. RAM iii. DMA iv. Cache memory.	5	1	3	3
	OR				
5. c)	Explain source-initiated asynchronous data transfer using handshaking method.	5	1	1	3
5. d)	Describe the structure of the magnetic disk with neat diagram.	5	2	2	3
6. a)	Explain RISC instruction pipeline in detail?	5	2	2	4
6. b)	Discuss about the following Interconnection structures i. Multiport memory ii. Hypercube system.	5	2	2	4
	OR				
6. c)	Explain about instruction pipeline in detail.	5	2	2	4
6. d)	With a neat diagram, explain parallel arbitration circuit.	5	1	2	4

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome